

EuroSAT Satellite Image Classification

Deep Learning Capstone Report - Final Experiment Version

Jianye Chen

Fanshawe College

PyTorch Image Classification Project

Final Model	Test Accuracy	Weighted F1-score
3-layer Custom CNN	92.07%	92.09%
Frozen ResNet18	82.44%	82.32%

Final choice: simple augmentation with restricted RandomResizedCrop and HorizontalFlip. VerticalFlip, Rotation pipeline, and ColorJitter were tested but not adopted because they did not improve the final model.

1. Project Overview

The objective of this capstone project is to build, train, evaluate, and compare deep learning models for image classification. The selected problem is satellite image land-use classification using the EuroSAT RGB dataset. The project uses PyTorch with standard image-classification components: Dataset/DataLoader, CNN layers, BatchNorm, Dropout, CrossEntropyLoss, Adam optimization, validation-based early stopping, and a frozen ResNet18 transfer-learning comparison.

The project compares a custom 3-layer convolutional neural network trained from scratch with a pretrained ResNet18 model used as a fixed feature extractor.

2. Dataset

EuroSAT contains 27,000 RGB satellite images across 10 land-use classes: AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, and SeaLake. A stratified split was used to preserve class proportions across the training, validation, and test sets.

Split	Number of Images	Purpose
Training	18,900	Model parameter learning
Validation	4,050	Hyperparameter selection and early stopping
Test	4,050	Final generalization evaluation

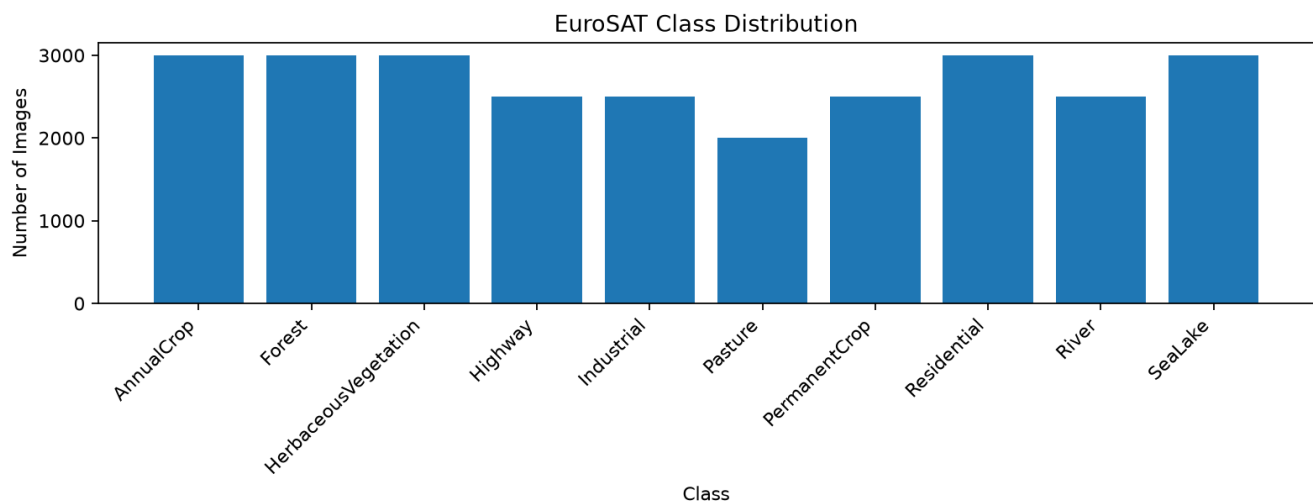


Figure 1. EuroSAT class distribution.

3. Preprocessing and Augmentation

All images were resized to 64 x 64 and normalized using ImageNet-style mean and standard deviation values. The final training transform intentionally stays simple: a restricted RandomResizedCrop with scale=(0.8, 1.0), RandomHorizontalFlip, ToTensor, and Normalize.

The restricted crop is important because satellite land-use labels often depend on global spatial structure such as field boundaries, road shape, river direction, or residential layout. A default random crop may remove too much of this structure.

Transform	Used in Final Model?	Reason
RandomResizedCrop(scale=(0.8, 1.0))	Yes	Preserves most of the satellite image while adding variation
RandomHorizontalFlip	Yes	Simple orientation variation
RandomVerticalFlip	No	Ablation decreased test accuracy
RandomRotation pipeline	No	Tested pipeline performed poorly and added extra preprocessing effects
ColorJitter	No	No clear improvement over baseline

4. Custom CNN Architecture

The custom model is a 3-layer CNN with three convolution blocks. Each block uses Conv2d, BatchNorm, ReLU, and MaxPool. After flattening, a fully connected layer with Dropout is used before the final 10-class output layer.

Stage	Operation
Block 1	Conv2d(3->32), BatchNorm, ReLU, MaxPool
Block 2	Conv2d(32->64), BatchNorm, ReLU, MaxPool
Block 3	Conv2d(64->128), BatchNorm, ReLU, MaxPool
Classifier	Flatten -> Linear(128*8*8 -> 256) -> Dropout -> Linear(256 -> 10)

5. Training Strategy

The model was trained using CrossEntropyLoss and Adam. Validation loss was monitored at the end of each epoch. When validation loss improved, the model state_dict was saved. If validation loss failed to improve for several epochs, early stopping was triggered and the best saved model was reloaded.

This prevents the final test result from depending on a later epoch that may have overfit the training data.

6. Hyperparameter Tuning

Three manual hyperparameter settings were compared. The final setting was selected using the lowest validation loss, not the highest training accuracy.

Experiment	Learning Rate	Batch Size	Dropout	Best Val Loss	Val Acc at Best Loss
Experiment 1	0.0010	32	0.30	0.4675	83.51%
Experiment 2	0.0005	32	0.30	0.4457	84.12%
Experiment 3	0.0010	64	0.50	0.6175	78.54%

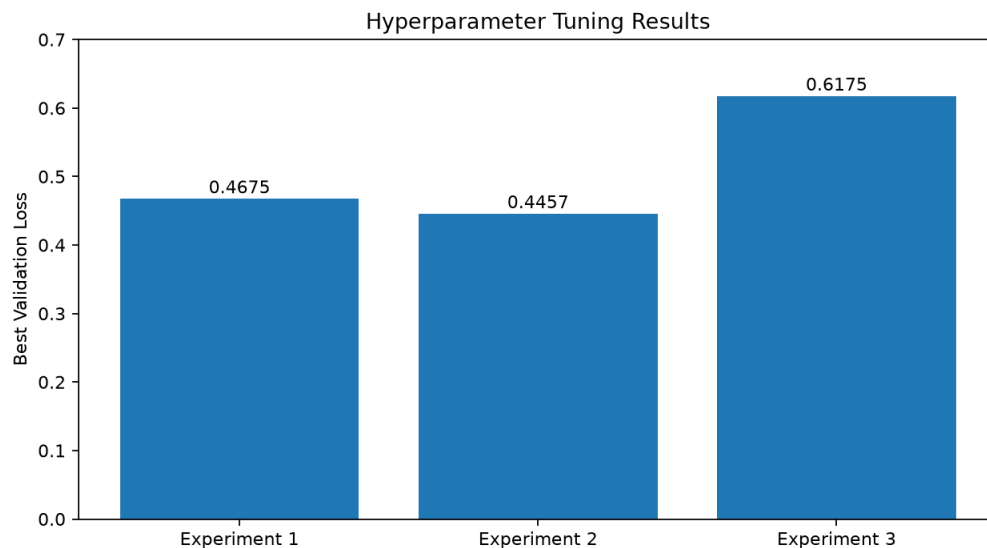


Figure 2. Hyperparameter tuning by validation loss. Experiment 2 was selected.

7. Baseline vs Incremental Improvement

The first baseline used a simple 3-layer CNN but only four tuning epochs and the default RandomResizedCrop. It achieved 85.60% test accuracy. The incremental version kept the same CNN architecture but changed only two training details: restricted crop scale and six tuning epochs. This improved test accuracy to 92.07%.

Version	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Baseline	84.37%	85.60%	85.89%	85.60%	85.49%
Incremental Final CNN	90.96%	92.07%	92.37%	92.07%	92.09%

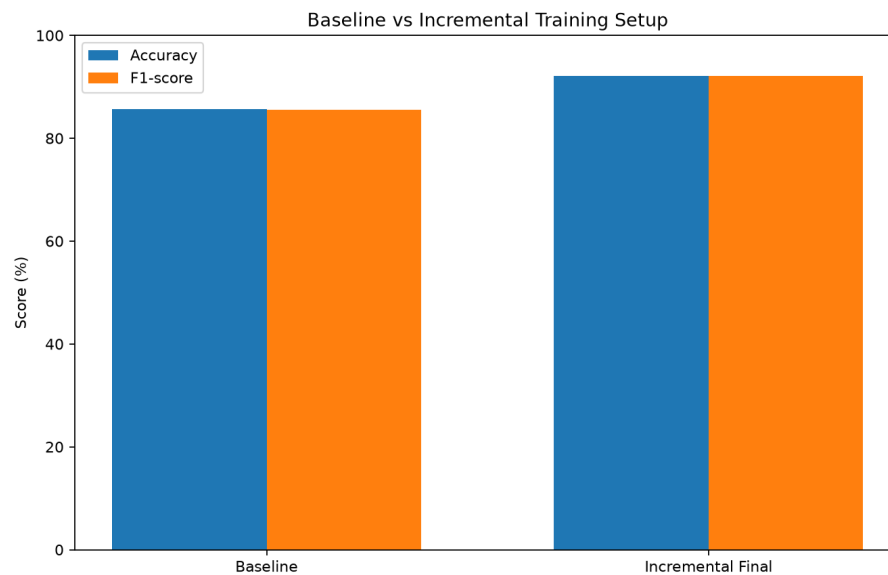


Figure 3. The improved training setup recovered strong performance without increasing architecture complexity.

8. Final Model Results

The final selected model is the incremental 3-layer CNN with learning rate 0.0005, batch size 32, dropout 0.3, restricted random crop, and horizontal flipping. It achieved strong test-set generalization.

Model	Validation Accuracy	Test Accuracy	Precision	Recall	F1-score
Final 3-layer CNN	90.96%	92.07%	92.37%	92.07%	92.09%
Frozen ResNet18	81.53%	82.44%	82.93%	82.44%	82.32%

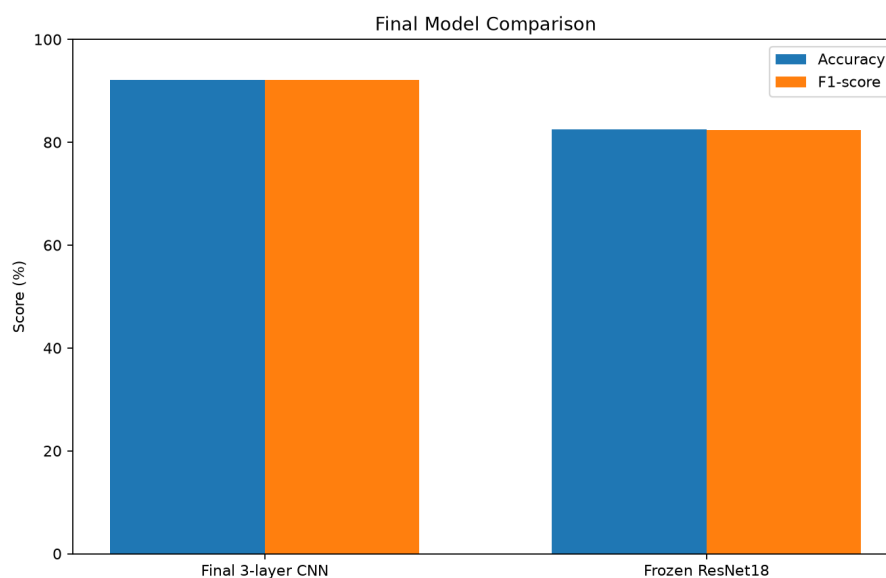


Figure 4. Final Custom CNN vs frozen ResNet18.

9. Confusion Matrix and Misclassification Analysis

The confusion matrix shows that the final model performs strongly across most classes. Remaining errors are mainly between visually similar land-use categories, such as agricultural and vegetation classes or linear structures such as rivers and highways.

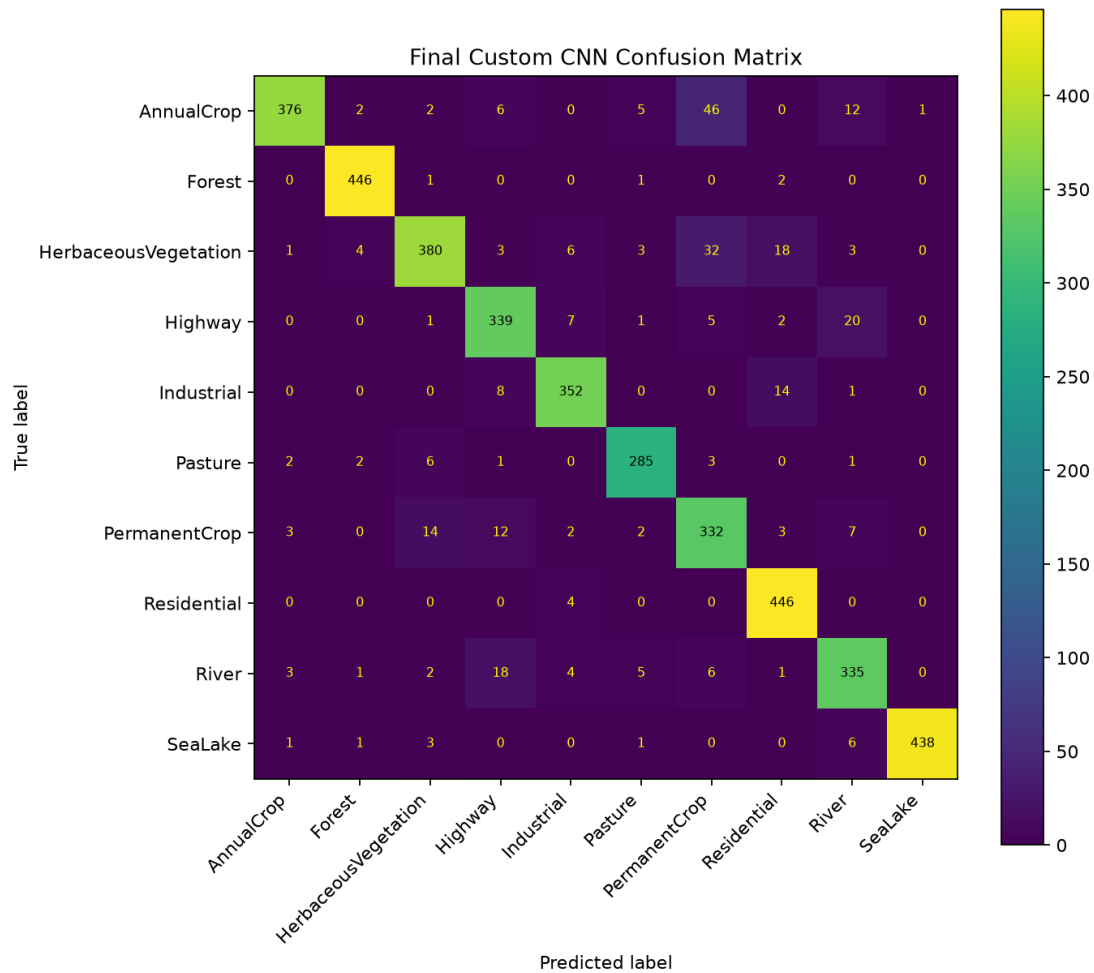


Figure 5. Final Custom CNN confusion matrix.

10. Ablation Study: Why Not Use Extra Augmentation?

Additional augmentation methods were tested one at a time while keeping the selected CNN architecture and hyperparameters fixed. None of the tested methods improved the final baseline.

Augmentation Setting	Test Accuracy	Weighted F1	Accuracy Change
Incremental Baseline	92.07%	92.09%	0.00 pp
+ ColorJitter	91.48%	91.49%	-0.59 pp
+ VerticalFlip	89.83%	89.95%	-2.25 pp
+ Rotation pipeline	67.60%	67.46%	-24.47 pp

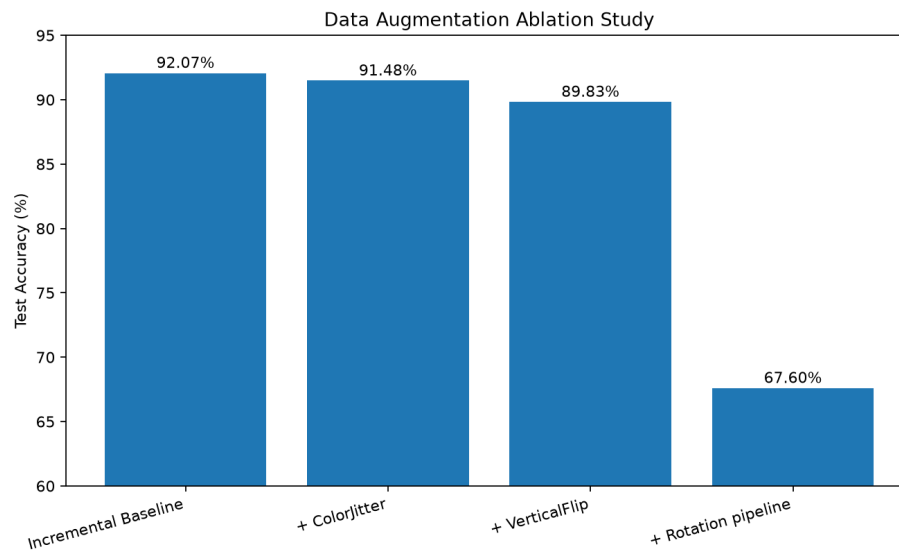


Figure 6. Additional augmentation did not improve the final model.

The rotation result is interpreted cautiously because the tested rotation pipeline also introduced resize-and-crop operations to avoid black borders. Therefore, the large decrease is not claimed to be the isolated effect of rotation alone. It only shows that this tested rotation preprocessing pipeline was not suitable for the final model.

11. Potential Problems and Mitigations

Potential Issue	Mitigation Used
Data leakage	Train, validation, and test sets were separated before model training. The test set was only used for final evaluation.
Overfitting	Dropout, BatchNorm, data augmentation, validation monitoring, and early stopping were used.
Misleading overall accuracy	Precision, recall, F1-score, and confusion matrices were also evaluated.
Overly aggressive augmentation	Ablation testing showed that more augmentation was not automatically better. The final model kept only the most effective augmentations.
Transfer-learning domain difference	ResNet18 was treated as a frozen-feature baseline. Future work could fine-tune later layers for satellite imagery.

12. Conclusion

This project built and evaluated a full PyTorch image-classification pipeline for EuroSAT. The final 3-layer CNN achieved 92.07% test accuracy and 92.09% weighted F1-score, outperforming the frozen ResNet18 comparison. The most important finding was that performance improved substantially by adjusting the training setup instead of increasing network complexity.

The ablation study also showed that extra preprocessing should not be added automatically. VerticalFlip, ColorJitter, and the tested rotation pipeline did not improve the final model, so the final solution uses the simplest tested augmentation pipeline that achieved the best result.

Future work could include fine-tuning the last ResNet blocks, testing a larger image size, using Grad-CAM to interpret the CNN decision process, and running multiple random seeds to measure training variability.